

**NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION
NORTHEAST FISHERIES SCIENCE CENTER
WOODS HOLE ECOSYSTEMS SURVEYS BRANCH
STATEMENT OF WORK
FOR THE
AUTONOMOUS UNDERWATER VEHICLE SCALLOP SURVEYS**

1.0 General

1.1 Background

The National Oceanic and Atmospheric Administration (NOAA), Fisheries, Northeast Fisheries Science Center (NEFSC) is the fisheries research arm of NOAA Fisheries in the region. The Center's Ecosystem Surveys Branch (ESB) develops and conducts fisheries-independent surveys that provide data to fisheries stock assessments and ecosystem monitoring programs. Traditional operational survey challenges and newly emerging human use activities have necessitated using additional novel sampling and monitoring methods to collect data in these areas for maintaining stock and ecosystem assessments.

1.2 Objective

This contract will support scallop fisheries-independent surveys that provide required data for the Atlantic sea scallop stock assessment by deploying autonomous underwater vehicles (AUVs) to collect optical data. In this request, we solicit a target of 200nm or more of optical AUV data acquisition using one or more AUVs within the Mid-Atlantic Bight, including designated scallop management areas. AUV operations and data acquisition are expected to occur between April 1 and August 31, 2026. The objective is for this data, in combination with the NEFSC's Habitat Camera (HabCam) Survey, to provide vital data needed for science and management of Atlantic sea scallops.

1.3 Scope of Work

This is a Non-Personal Service contract; the contractor shall provide all plant, trained labor, equipment, materials, AUV supervision, transportation, permits, insurance, and data quality control necessary to provide for AUV optical survey data for the NEFSC ESB. Performance under this contract is known as "Work". The contractor shall be fully knowledgeable of all contract requirements and shall ensure Work is accomplished in accordance with the terms and conditions of the contract in a manner that will promote and maintain a safe environment.

The contractor shall be responsible for all logistical components of conducting a 200nm survey using AUVs with optical payloads in the Mid-Atlantic Bight (unless specified otherwise).

1.1.1 The operator shall be a fully trained and authorized employee of the company that owns the AUVs and shall be fully knowledgeable of all contract requirements. The contractor shall at all times ensure work is accomplished in accordance with the terms and conditions of the contract in a safe and professional manner. The contractor shall have demonstrated experience with AUVs, knowledge of NEFSC AUV-scallop efforts, and experience with NEFSC equipment required for integration with the AUVs.

1.1.2 Special Qualifications and Permits. The contractor shall ensure all required skill sets, and permits are obtained and maintained for performance under this contract.

1.1.3 Quality Control. Quality Control is the responsibility of the Contractor. In the event that data is transmitted to shore, the contractor shall provide the data updates to the government so that the quality of the data can be established during the survey. All data will be provided from the Contractor to NEFSC ESB at the end of the survey.

2.0 Safety, Security, Fire Protection, Environmental Controls, Conservation of Utilities, and Compliance with Laws and Regulations.

2.1 Safety Requirements. The contractor shall take all necessary precautions to meet proper safety standards, and regulations of all local, state and federal codes and regulations.

3.0 Government Responsibilities:

3.1 Government Furnished Property, Materials, Equipment and Services (GFP/M/E/S). The government will provide overall scientific guidance for the surveys, such as survey design, trackline waypoints in a digital format amenable to the AUV operators, survey timing, and survey location. As needed, the government shall provide encrypted portable hard drives (certified by NEFSC IT) for data transfer from the AUVs. The Government shall provide up to two optical payloads (Autonomous Benthic Imaging Systems) to be used in the survey to ensure the data are of high quality for the sea scallop stock assessment and compatible with the NEFSC ESB's HabCam Survey data pipelines. The Government shall work with the Contractor to identify options and resources in the case of an emergency situation with the AUVs.

4.0 Contractor Responsibilities.

4.0 Contractor Furnished Property, Materials, Equipment and Services (GFP/M/E/S). The contractor shall provide all plant, trained labor, equipment, materials, supervision, transportation, permits, insurance, and quality control necessary to achieve the quality performance standards of the work in this contract. Scientific instrumentation provided by the contractor includes the AUVs and required supplies for their operations, and data output in a format that is compatible with current HabCam Survey data.

4.1 Contractor shall immediately notify the Contracting Officer Representative (COR) if discrepancies are discovered between the existing property conditions and those noted on the Performance Work Statement (PWS).

4.2 The contractor shall notify the COR immediately when it is discovered that the deliverable schedule will not be met to work together to revise the schedule as necessary. The contractor shall communicate with the Contracting Officer (CO) and revise the schedule as necessary. The contractor shall not deviate from schedule without prior approval from the CO.

4.3 The contractor shall maintain an adequate workforce trained to safely and satisfactorily perform under this contract.

4.4 The contractor shall ensure all employees and subcontractors are fully trained and have a full understanding of the statement of work task requirements; the government will not provide instruction or oversight.

4.5 Contractor shall ensure all Contractor employees, and all associated subcontractor(s) are experienced in the type of work involved and familiar with the specifications of this contract. Contractor is responsible for damage to any Government equipment and supplies caused by Contractor personnel.

5.0 Deliverables.

1. Requirements

- Payload integration - The AUV(s) must be capable of using the NEFSC's optical payload, known as the Autonomous Benthic Imaging System (ABIS), to collect the optical scallop data. ABISs are equipped with the cameras, sensors, configuration, and software compatible with the HabCam vehicle, allowing for data integration between the platforms to be seamless. As such, Long-Range AUVs must be used for the survey.
- ABIS calibration - The contractor will be responsible for leading the ABIS camera calibrations with their vehicles. NEFSC staff will assist using protocols developed for the HabCam survey, but the contractor will be responsible for ensuring the proper facilities (e.g. dunk tanks) are committed for calibrations.
- Autonomy - The AUV(s) shall be able to navigate independently, sufficient to the objectives of the scallop survey, without requiring additional aids that require the deployment of extra equipment (such as long or short baseline acoustics).
- Tracklines - The AUV(s) must collectively be able to collect at minimum 200nm of optical scallop data. Final tracklines to be sampled in the Mid-Atlantic Bight will be determined by NEFSC personnel after the award based on the number of AUVs to be used, HabCam Survey logistics, and sea scallop stock assessment needs.
- Speed – The AUV(s) must be capable of maintaining a consistent survey speed. Required survey speeds will be 1-2 kts.
- Endurance – The AUV(s) must be capable of operating for 3 days or longer on one battery charge to minimize recovery and deployments for refueling or data transfer.
- Depth - The AUV(s) must be able to hold a stable distance from the sea bed of 2 meters or less while traversing and collecting optical data, and be able to operate up to 200m of water depth.
- Emergency Behaviors - The AUV(s) should have emergency behaviors, such as loitering, automatic surfacing, or sea-bed parking, to ensure vehicle and data recovery. It is preferred that the AUV(s) equipped with emergency recovery beacons or pingers that is independently powered is preferred for the AUV
- Deployments - The AUV(s) should be deployable and recoverable over the side by a ship's crane without ramps or other very large and specialized handling equipment.
- Communications – The AUV(s) must be able to communicate with the contractor (through means such as satellite, cellular and Wi-Fi) while underway to ensure vehicles are operating, data is being collected, and missions can be revised as needed. Use of a dashboard during the missions by the contractor and for NEFSC viewership is preferable to understand vehicle positioning and status.
- Deployment and Recovery – A system to deploy and recover the USV from dockside facilities in a safe and efficient manner.
- Insurance - The contractor will be responsible for the loss of equipment during the survey.

2. Communication and Deliverables

- The contractor will meet with the NEFSC to discuss the proposed AUV mission(s) based on the scallop assessment needs and timelines (work must be completed by August 31, 2026). The contractor is expected to communicate with the NEFSC throughout the contract period (through in-person, virtual, or telecommunication) to ensure the project meets the needed timelines.
- Data Delivery - appropriate metadata (positioning, vehicle state et.c), oceanographic data, and imagery data from the contractor will be provided to the NEFSC within one (1) week of the mission being completed. Images do not need to be annotated by the contractor; NEFSC will analyze imagery data post-mission.
- Report - The contractor will provide a final report summarizing the AUV mission(s), with relevant maps and plots, as well as mission summary statistics (e.g. distance traveled with image data collected and total distance traveled, start and end dates of the AUV mission(s), battery usage (absolute and percent of total), mission duration (e.g. decimal days). The report will be due within two (2) months of completing the work.

6.0 Deliverables.

The contractor shall be fully knowledgeable and skilled in the scientific disciplines required to successfully fulfill this requirement. Contractor shall perform the following tasks and provide the following deliverables:

- **Task 1 Description (CLIN 0001):** Pre-mission communication and preparation for the AUV(s).
- **Task 1 Deliverable (CLIN 0001):** Kickoff meeting between NEFSC personnel and the contractor. AUV(s) preparation and integration with ABIS(s). Complete camera calibrations. Equipment preparation for travel to AUV(s) launch location.
- **Task 2 Description (CLIN 0002):** Conduct AUV optical scallop survey mission(s).
- **Task 2 Deliverable (CLIN 0002):** Mobilization to appropriate launch location that aligns with meeting the sea scallop stock assessment needs. Launching of AUV(s) via the best method as identified by the contractor (small boat, large vessel, shore launch) and ensure all systems are functional prior to AUV deployment(s). Conduct AUV mission(s) with optical sea floor data being acquired. Retrieve AUV(s), ABIS(s), and data. Return to Contractor site for demobilization and data offload.
- **Task 3 Description (CLIN 0003):** Data delivery.
- **Task 3 Deliverable (CLIN 0003):** Deliver data to the NEFSC.
- **Task 4 Description (CLIN 0004):** Documentation and closeout.
- **Task 4 Deliverable (CLIN 0004):** Provide a final report to the NEFSC including required materials. Conduct a closeout meeting between the contractor and the NEFSC as guided by the NEFSC.

7.0 Place(s) of Performance

Northeast Fisheries Science Center 166 Water St, Woods Hole, MA 02543 or Vendors Location
Mid-Atlantic Bight